

Project Proposal

RESNOR

1. Executive Summary

RESNOR is an intelligent, data-driven academic monitoring platform designed to transform the way students and educators engage with learning. Built to integrate seamlessly with existing university platforms such as DIU's BLC and AI Professor, RESNOR goes beyond simple completion tracking by offering deep behavioral analytics, predictive academic modeling, personalized AI tutoring, and proactive well-being support.

This proposal outlines the problem RESNOR addresses, its core objectives, system architecture, key features, innovation highlights, and the tangible impact it is expected to deliver for students, educators, and institutions.

2. Problem Statement

Despite the growing adoption of digital learning platforms in universities, a critical gap persists between the data these platforms collect and the actionable insights they provide. Specifically:

- Students lack personalized, real-time feedback on their learning progress, making it difficult to self-identify weaknesses or adjust study strategies proactively.
- Existing platforms such as BLC/AI Professor offer only surface-level indicators (Done / To Do), with no predictive or diagnostic capabilities.
- Educators have limited visibility into individual student struggles, preventing early intervention for at-risk learners.
- The strong relationship between student mental well-being and academic performance is consistently overlooked by current systems, leaving students without timely, non-intrusive academic support.

These gaps result in declining engagement, preventable academic failure, and missed opportunities for personalized growth. RESNOR is designed to close these gaps comprehensively.

3. Project Objectives

RESNOR is guided by the following core objectives:

- Deliver a real-time, visual representation of each student's academic progress and engagement.
- Identify weak, incomplete, or neglected topics automatically through AI-driven behavioral analysis.
- Provide an AI-powered tutoring assistant capable of personalized academic guidance.
- Predict academic outcomes such as CGPA to help students plan and adjust study strategies proactively.
- Promote consistent study habits through gamification and intelligent notification systems.
- Equip educators with data-driven dashboards for monitoring class-wide performance and identifying at-risk students.

- Support student well-being by detecting signs of academic stress and providing lightweight, ethical AI-based encouragement.

4. Proposed Solution & System Overview

RESNOR connects to existing university LMS platforms (BLC / AI Professor) and ingests student activity data — including completed materials, pending tasks, time-on-content, and quiz performance. This raw data is processed through a multi-layer AI and analytics pipeline to generate insights, predictions, and personalized recommendations delivered to both students and teachers via dedicated dashboards.

4.1 System Architecture

The platform is structured across four functional layers:

- **Data Source Layer:** BLC / AI Professor LMS platforms serve as the primary data source, providing raw student activity logs, material completion records, and assessment results.
- **Data Processing Layer:** Incoming data is cleaned, normalized, and structured for downstream analysis, ensuring consistency and reliability.
- **AI & Analytics Layer:** Core intelligence modules including CGPA prediction, engagement analysis, quiz generation, Digital Twin simulation, and well-being detection operate on the processed data.
- **Application Layer:** Clean, intuitive dashboards for students and teachers, plus an AI Tutor chatbot and a smart notification engine, present insights in actionable form.

5. Key Features

RESNOR comprises eleven integrated features, spanning academic analytics, AI assistance, and well-being support:

Feature	Description
Student Growth Dashboard	Visualizes completed vs. pending materials, topic-wise performance, and long-term growth trends using interactive graphs and analytics.
AI Tutor	A conversational academic assistant that explains concepts, answers queries, and provides simplified guidance grounded in official BLC/AI Professor course content.
AI Quiz Generator	Automatically generates assessments from study materials and tracks accuracy, improvement trajectories, and weak areas over time.
CGPA Prediction System	Forecasts future CGPA based on current performance data and study consistency, enabling proactive academic planning.
Digital Twin	Builds a virtual model of each student to simulate how behavioral changes — such as increased study hours — would affect academic outcomes.

"Explain My Mistake" AI	Analyzes incorrect quiz responses to identify the underlying conceptual misunderstanding and delivers targeted corrective explanations.
Passive Learning Detection	Distinguishes between genuine engagement and surface-level activity (e.g., opening materials without meaningful interaction) to surface true learning behavior.
Gamified Progress System	Encourages consistent habits through study streaks, achievement badges, and engagement scores — rewarding behavioral improvement over raw scores.
Smart Notification Engine	Sends personalized timely reminders when progress stalls or topics remain unaddressed, reducing procrastination and supporting continuity.
Teacher Dashboard	Provides educators with class-wide and individual-level performance analytics, identifies struggling students, and enables targeted AI-assisted interventions.
AI Well-being Support	Monitors behavioral indicators of academic stress (sudden activity drops, declining performance) and sends friendly, non-intrusive motivational check-ins and study-balance suggestions.

6. Innovation & Differentiators

RESNOR introduces several capabilities that distinguish it from existing academic platforms:

- **Digital Twin Simulation:** No existing university LMS offers a virtual behavioral model that lets students explore "what-if" academic scenarios before committing to a study strategy.
- **"Explain My Mistake" AI:** Moving beyond score reporting, this feature performs root-cause analysis on incorrect answers to address conceptual gaps directly.
- **Integrated Well-being Awareness:** Academic performance data is combined with engagement signals to identify stress and burnout early — a capability absent from standard LMS platforms — while maintaining strict ethical boundaries (non-diagnostic, privacy-preserving).
- **Passive vs. Active Learning Discrimination:** The system distinguishes between opening content and meaningfully engaging with it, providing a more accurate picture of actual student effort.
- **Predictive CGPA Modeling:** Continuous, data-driven CGPA forecasting gives students and advisors an early-warning system for academic risk.

7. Ethical Considerations

RESNOR is designed with responsible AI principles at its core:

- The well-being support module is explicitly non-diagnostic and does not attempt to replace professional psychological services. It provides only general, academically-focused encouragement and study guidance.
- All student data is handled in accordance with institutional data privacy policies. No behavioral or well-being data is shared with third parties.

- Gamification mechanics are designed to promote healthy study habits, not to create competitive pressure or anxiety.
- Predictive models are presented as guidance tools, not deterministic verdicts, empowering students to make informed decisions rather than fostering fatalism.

8. Expected Impact & Outcomes

Successful deployment of RESNOR is expected to deliver the following outcomes:

- Measurable improvement in student academic performance through personalized, data-driven guidance.
- Increased student engagement and study consistency, evidenced by gamification metrics and activity trends.
- Earlier detection of at-risk students, enabling timely educator intervention and reducing dropout risk.
- More efficient and targeted teaching, with educators empowered by actionable class-wide analytics.
- Improved student well-being and reduced academic stress through proactive, non-intrusive support mechanisms.
- A scalable, institution-agnostic platform architecture capable of integration across multiple university LMS environments.

9. Conclusion

RESNOR represents a fundamental shift in how academic progress is understood, monitored, and supported. By combining real-time behavioral analytics, predictive AI modeling, personalized tutoring, and ethically grounded well-being awareness into a single, integrated platform, RESNOR transforms passive data into active, student-centered intelligence.

This project does not merely track what students have done — it anticipates where they are headed, explains why they struggle, and provides the right support at the right moment. We believe RESNOR has the potential to redefine the standard for intelligent academic support systems in higher education, and we are excited to present it for consideration in this competition.